Vathaluru Akhileswar Reddy

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Profile Summary

- Backend engineer specializing in building production-grade systems with FastAPI and Spring Boot, featuring domain-driven design, microservices architecture, and serverless AWS deployments.
- Expert in AI agent development using LangGraph and LangChain for conversational AI with multi-turn dialogue management, custom tool integration, and vector databases.
- Proven track record of designing scalable architectures including event-driven systems with RabbitMQ, RESTful APIs with OAuth2/JWT authentication, and CI/CD pipelines achieving 80%+ test coverage.

Technical Skills

Languages: Python, Go, Java, JavaScript, C, C++
Backend Frameworks: FastAPI, Spring Boot, Express.js
AI/LLM: LangGraph, LangChain, OpenAI API, Tavily, Inngest Agent Kit
Frontend: React, Tailwind CSS, Vite
Microservices: Spring Cloud (Gateway, Eureka, Config), RabbitMQ
Databases: PostgreSQL, MongoDB, MySQL, Redis, Pinecone (Vector DB)
Cloud & DevOps: AWS (Lambda, S3, SES, Cognito), AWS CDK, Docker, GitHub Actions
ORM: SQLAlchemy, Spring Data JPA, Mongoose
Auth & Payments: OAuth2, JWT, AWS Cognito, Clerk, Razorpay
Testing: Pytest, Ruff, Mypy, JUnit

Education

KL University — **Guntur, India**
Bachelor of Technology in Computer Science and Engineering, CGPA: 9.43 — *2021–2025*

Experience

Ideyalabs — **Hyderabad, India**
Backend Developer — *2025–Present*

APEX – Transportation Back Office System

- Architected a FastAPI backend with domain-driven design supporting multiple core domains (Clients, Pricing & Rating, Transit Times, Billing) with clean separation between API, service, and data layers.
- Deployed a serverless application on AWS Lambda using AWS CDK for infrastructure-as-code, with CI/CD pipelines via GitHub Actions achieving 80%+ test coverage.
- Designed and implemented comprehensive RESTful APIs with JWT authentication via AWS Cognito, input validation, and structured error handling.
- Engineered PostgreSQL persistence using SQLAlchemy ORM with a multi-schema architecture.
- Integrated multiple external services: SMC3 (freight rating), HERE.com (geocoding), EIA (fuel pricing), AWS S3 (storage), and AWS SES (email).
- Implemented Excel/CSV export generation, PDF generation for quotes, and structured JSON logging for API monitoring.

Ideyalabs — **Hyderabad, India**
Backend Developer Intern — *2024–2025*

HLF – Blockchain Transaction System

- Implemented RESTful APIs in Go for blockchain transaction management including issue, redeem, and transfer operations.
- Engineered Go-based smart contracts (chaincode) for Hyperledger Fabric to enforce business logic across multiple workflows.
- Led migration from Corda-based blockchain to Hyperledger Fabric, improving performance and interoperability.

Projects

Chatbot:

- Engineered a full-stack AI chat application with React + Tailwind CSS frontend and FastAPI backend supporting multi-threaded conversations with PostgreSQL-backed persistent history.
- Integrated OpenAI GPT-4 with LangGraph for stateful conversation management and custom tools, including Tavily web search for real-time information retrieval.
- Designed a service-layer architecture using SQLAlchemy ORM with proper session management for handling concurrent conversations.

Budget Buddy:

- Built an AI-powered command-line finance assistant using LangGraph and LangChain with intelligent financial tracking capabilities and a conversational AI interface.
- Created custom financial tools supporting income/expense tracking with date range analytics, real-time balance monitoring, and web search integration.
- Implemented a modular architecture with PostgreSQL, SQLAlchemy ORM, dependency injection, and persistent conversation memory.

Fitness AI Platform:

- Designed a microservices platform using Spring Boot with service discovery (Eureka), API gateway (Spring Cloud Gateway), and centralized configuration management.
- Implemented an event-driven architecture using RabbitMQ for asynchronous inter-service communication between user, activity, and AI recommendation services.
- Built an AI recommendation service using Google Gemini API with Spring WebFlux for reactive programming, MongoDB storage, and OAuth2 authentication.

ClearBG:

- Implemented a full-stack image background removal application using Spring Boot backend, React frontend, and ClipDrop API integration.
- Integrated Clerk authentication with webhook synchronization for user management and JWT-based API security using custom filters.
- Added a credit-based payment system using Razorpay integration and reactive Spring WebClient for asynchronous AI service communication.

Travel Agent AI:

- Developed an AI travel assistant using LangGraph with FastAPI backend, providing intelligent flight and hotel search capabilities through natural language queries.
- Integrated Pinecone vector database for conversation context retrieval and implemented custom tools for SerpAPI-based search with structured JSON responses.
- Designed an agent-based architecture with conversation summarization, PostgreSQL for data persistence, and LangChain for tool orchestration.

For more projects, visit: github.com/Akhil351

Achievements

LeetCode: Solved 530+ problems across Easy, Medium, and Hard difficulty levels covering arrays, trees, graphs, recursion, and dynamic programming.

Red Hat Certified: Enterprise Application Developer (EX183) – Validated expertise in Java EE application development, RESTful services, CDI, JPA, and JBoss EAP deployment.

Hackathon Finalist: Google Cloud Agentic AI Day (Hack2skill, 2025) – Ranked in the top 700 out of 57,000+ participants. Built an AI-powered monitoring system using Vertex AI, Gemini, Next.js, and ESP32 hardware within a 30-hour timeframe.